Cover Letter — Submission to Remote Sensing of Environment

Supranab Panda Center for Environment and Climate Institute of Technical Education and Research Siksha ‘O’ Anusandhan (Deemed to be) University Bhubaneswar 751030, Odisha, India

[Date of submission]

The Editors-in-Chief *Remote Sensing of Environment* Elsevier B.V.

Dear Editors,

I am pleased to submit our original research article entitled **“Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval: A Multi-Year Bay-of-Bengal Coastal Framework (2017–2024)”** for consideration in *Remote Sensing of Environment*.

Why this manuscript matters now. Almost every published Sentinel-1 rice phenology algorithm uses the SAR backscatter trough at flooding/transplanting as the anchor for start-of-season detection. Yet in cyclone-prone Asian deltas — which produce roughly 11 % of global rice — that same trough can be generated by storm-surge saline inundation weeks earlier in the season, silently corrupting derived phenological dates in precisely the years that vulnerability assessments need most. To our knowledge, no prior study has characterised, let alone corrected, this physical confound.

What we did. Using Cyclones Fani (May 2019), Bulbul (November 2019), Amphan (May 2020) and Yaas (May 2021) as natural experiments in a Before-After-Control-Impact design across five coastal Odisha districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) with three inland controls (Dhenkanal, Angul, Cuttack), we (i) developed an open-source random-forest classifier that separates cyclone-induced saline inundation from agronomic transplanting flooding using seven Sentinel-1, Sentinel-2, JRC water, and ERA5 features (full list in Supplementary Note S3 §S3.5); (ii) propagated the classifier output into a parallel raw vs. corrected phenology-extraction pipeline based on Whittaker-smoothed, double-logistic curve fitting; and (iii) quantified the resulting bias in SOS/POS/EOS dates with a mixed-effects BACI model operationalised as a two-way fixed-effects difference-in-differences (TWFE-DiD) estimator with district-clustered inference. We validated using a fully open-data, multi-source strategy: against the NASA MODIS MCD12Q2 Land Surface Phenology product as the methodologically independent primary reference, against ICRISAT Village Dynamics in South Asia (VDSA) microdata for the Bhadrak benchmark site, against district-level Kharif rice yield records from data.gov.in, against Sentinel-2 10-m visual interpretation at sixty stratified sites (the freely-redistributable fallback to PlanetScope NICFI, activated under the pre-registered §E5 path after the Tropical Forest Observatory programme restricted eligibility to forest-domain users), and against two existing rice mask products (Mondal et al. 2022; Singha et al. 2019). We additionally tested transferability on the Andhra Pradesh coast under Cyclone Hudhud (October 2014). Every dataset used in the study is publicly downloadable without permission, application, or institutional gatekeeping — a deliberate design choice that maximises reviewer reproducibility and removes any data-access dependency from the publication record.

Headline empirical findings. The retrained random-forest saline-flood classifier achieves overall accuracy of 0.990 (F1 = 0.990; 5-fold CV OA = 0.996; n = 96 stratified hold-out, n = 480 total labels) on the full Sentinel-1 + Sentinel-2 + ERA5 feature set, and OA = 0.844 on a SAR-only robustness variant that drops the cloud-affected optical features. Applying the classifier as a district-aggregated cyclone-flood pixel-share mask to the BACI phenology panel produces a small but measurable attenuation of the DiD coefficient (τ_SOS: +15.289 → +15.108 days; τ_EOS moves from a degenerate 0.000 to −0.239 days). The small magnitude reflects the bounded cyclone-flood pixel share at the district scale (Fani 0.04–2.5%, Amphan 0.005–1.9%, Yaas 0.018–7.2% per district), confirming that the surge confound is real and detectable at the pixel scale but partially diluted at the district-aggregation scale. The pre-registered direction τ_raw > τ_corrected > 0 is empirically confirmed for SOS, with the WCR-restricted 95% CI inclusive of zero — reported as a transparent null with confirmed bounded attenuation rather than over-claimed effect. All five robustness instruments (wild-cluster restricted bootstrap, leave-one-out jackknife, in-space and in-time placebo, post-hoc MDE) converge consistently on the corrected panel.

Why RSE specifically. This work fits the journal's scope on biophysical applications of remote sensing to agriculture in three ways: (1) it removes a previously uncharacterised barrier to robust phenological retrieval rather than incrementally improving accuracy; (2) it delivers a named, openly-licensed deployable artefact (the RiceBaCI-GEE toolkit at https://github.com/pandasupranab/RiceBaCI-GEE and a 10-m corrected SOS/POS/EOS dataset on Mendeley Data, DOI 10.17632/z3zxk4xy3c.1); and (3) it speaks directly to the climate-vulnerability and parametric-insurance communities that the *RSE* readership serves. The methodology was pre-registered on the Open Science Framework prior to data analysis (https://osf.io/c4mp8 (DOI: 10.17605/OSF.IO/C4MP8)), and all code is publicly available under MIT licence.

Originality and ethics. The manuscript has not been published elsewhere and is not under consideration at another journal. All authors approve this submission and have made substantial intellectual contributions consistent with the journal's authorship policy. We declare no competing interests, no funding to disclose, and we report the use of generative-AI assistants in the manuscript as required by Elsevier policy.

Suggested reviewers. We propose the following experts whose recent work most directly bears on our contribution; none is a current or recent collaborator:

1. **Dr. Mirco Boschetti** — CNR-IREA, Milan, Italy (mirco.boschetti@irea.cnr.it). Lead author of the PhenoRice algorithm — the most cited rice phenology retrieval framework; his expertise is directly relevant to our SOS/POS/EOS double-logistic methodology.

2. **Dr. Mrinal Singha** — Indian Institute for Human Settlements, Bengaluru, India (mrinal.singha@iihs.ac.in). Author of the South-Asia Sentinel-1 paddy rice product (Singha et al. 2019, *Scientific Data*) and the Bangladesh flood-affected rice paper (Singha et al. 2020, *ISPRS J P&RS*) — both cited in our manuscript.

3. **Dr. Nguyen Lam-Dao** — Vietnam National Space Center, VAST, Ho Chi Minh City, Vietnam (nldao@vnsc.org.vn). Lead author of the Mekong-Delta Sentinel-1 salinity-rice mapping (Hoang-Phi et al. 2020, *Remote Sensing*) — directly comparable saline-inundation-rice problem in a different delta, providing methodological cross-check.

4. **Dr. Saptarshi Mondal** — VassarLabs / Birla Institute of Technology Mesra, India (saptarshi.mondal@vassarlabs.com). Author of the Odisha rice-fallow Sentinel-1 work (Chandna and Mondal 2020, *GIScience & Remote Sensing*) — eastern Indian rice remote-sensing expertise on the same study area.

5. **Prof. Clement Atzberger** — University of Natural Resources and Life Sciences (BOKU), Vienna, Austria (clement.atzberger@boku.ac.at). Senior authority on agricultural remote sensing time series; *RSE* contributor with broad methodological expertise.

We have no reviewers to exclude due to conflict of interest.

Reproducibility statement. The submitted manuscript corresponds to GitHub release v1.0.1-submission, permanently archived on Zenodo (this-version DOI 10.5281/zenodo.20587316; concept DOI 10.5281/zenodo.20024578). The accompanying derived data deposit (corrected SOS/POS/EOS rasters, district BACI panel, classifier label set, RF model card v0.3.0, SHA256 checksums) is published on Mendeley Data under CC-BY-4.0 (DOI 10.17632/z3zxk4xy3c.1). Every numerical result in the manuscript and supplement is computed from real public Earth-observation data: Copernicus EMS rapid-mapping (Fani EMSR348), Sentinel-1 GRD, Sentinel-2 L2A (via Microsoft Planetary Computer STAC for the Bulbul transferability probe; via Google Earth Engine for the main BACI panel), JRC Global Surface Water v1.4, ECMWF ERA5, IBTrACS NI v04r01, and GADM v4.1. All headline coefficients, all five robustness instruments (wild-cluster bootstrap, leave-one-out jackknife, in-space and in-time placebo, post-hoc MDE / power, and the pre-registered Cyclone Bulbul out-of-sample transferability probe), all figures, and all supplementary tables S1–S10 and S13a are computed on real inputs at v2.1 of the corrected pipeline. The Bulbul probe (Supplementary Note S1 / Table S3) returns a mean residual of $+6.56\,d$ with 4 of 4 paddy-dominant probe districts inside the 95% prediction interval, satisfying the pre-registered pass criterion.

I thank the Editors and reviewers in advance for their consideration of our manuscript.

Yours sincerely,

Supranab Panda PhD Researcher, Agricultural Meteorology Center for Environment and Climate Siksha ‘O’ Anusandhan (Deemed to be) University, Bhubaneswar, Odisha, India ORCID: 0009-0009-6496-6545 Email: pandasupranab@gmail.com GitHub: https://github.com/pandasupranab

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Pre-submission self-check.

1. Confirm the OSF and GitHub URLs are live and public.
2. Mendeley Data DOI minted and published: 10.17632/z3zxk4xy3c.1 (8 June 2026).

3. Have your supervisor read the cover letter; their initials at the bottom strengthen credibility.
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